

NightClean

Smart Cities Deserve Smart Cleaning

An AI-powered autonomous garbage-cleaning system that operates during night hours to keep cities spotless by morning. Combining computer vision, robotics, and citizen engagement for next-generation urban waste management.

The Urban Waste Crisis



The Problem

Cities worldwide face mounting challenges with street waste management. Manual cleaning crews struggle to keep up with daily waste generation, while daytime operations create traffic congestion and road safety concerns. Without systematic monitoring and reporting mechanisms, municipalities lack visibility into problem areas.

Current waste collection systems are reactive rather than proactive, responding to complaints rather than preventing accumulation. This results in persistent dirty spots, public health hazards, and frustrated citizens.

Why This Matters Now

Public Health

Accumulated waste harbors bacteria, attracts pests, and creates air pollution through decomposition

Urban Environment

Litter in streets washes into drains during rain, causing blockages and water contamination

Smart City Vision

Modern cities need intelligent, automated systems to handle infrastructure maintenance efficiently

With urban populations growing rapidly, traditional waste management approaches cannot scale effectively. Cities need autonomous, data-driven solutions that optimize resource allocation and ensure consistent cleanliness across all neighborhoods.

Introducing NightClean

1

Autonomous Night Cleaning

Self-driving trucks patrol empty roads between 12 AM–5 AM, collecting waste without disrupting daytime traffic

2

AI Vision Detection

Computer vision identifies garbage in real-time using trained machine learning models

3

Citizen Reporting

Residents upload photos of problem areas through web/mobile platform for targeted response



System Architecture

Data Sources

Sensors, cameras, and citizen reports collected

Cloud Ingest

Secure upload and preprocessing in cloud

AI Processing

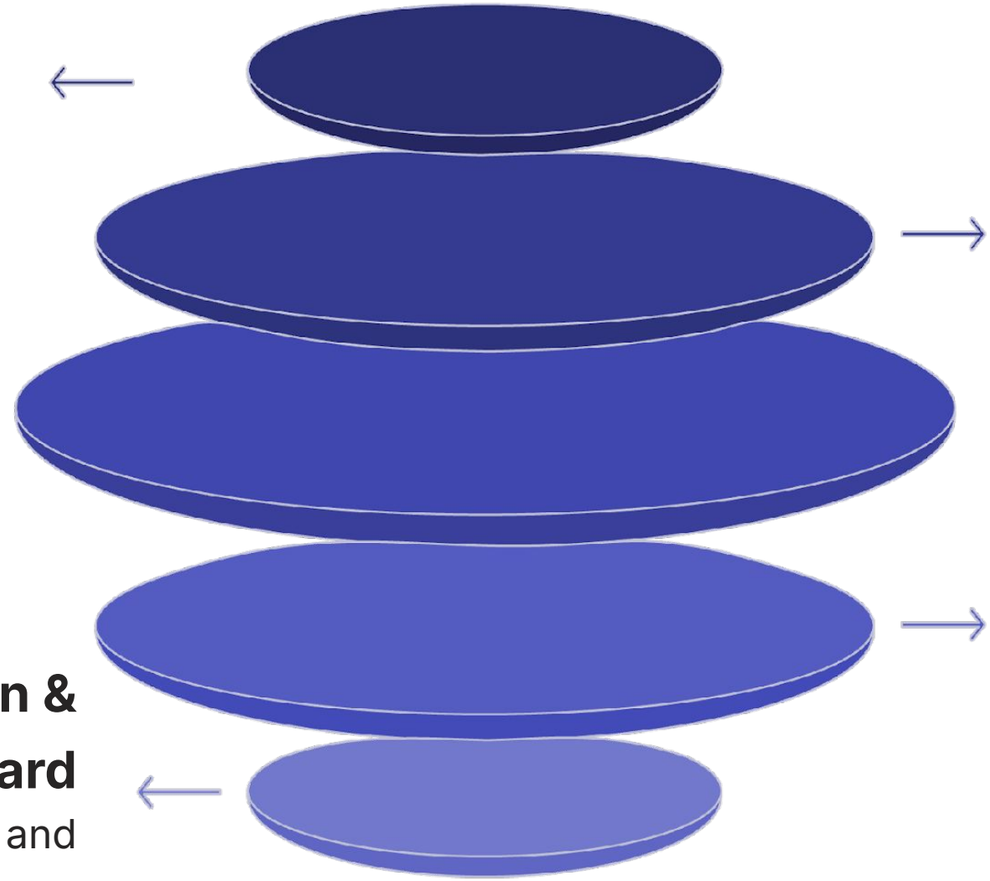
Computer vision and analytics applied

Route Optimization

Compute optimal paths and controls

Actuation & Dashboard

Navigation updates and dashboard sync



Hardware Components



Processing Unit

Raspberry Pi or ESP32 microcontroller serving as the brain, processing sensor data and executing control commands in real-time



Vision System

High-resolution camera module captures street imagery continuously, feeding frames to AI detection models



Distance Sensors

Ultrasonic sensors detect obstacles and measure distances, ensuring safe navigation around objects



GPS Module

Precise location tracking enables accurate mapping of waste locations and optimized route planning



Motor Drivers

Control wheel motors for autonomous movement and robotic arm/suction system for waste collection



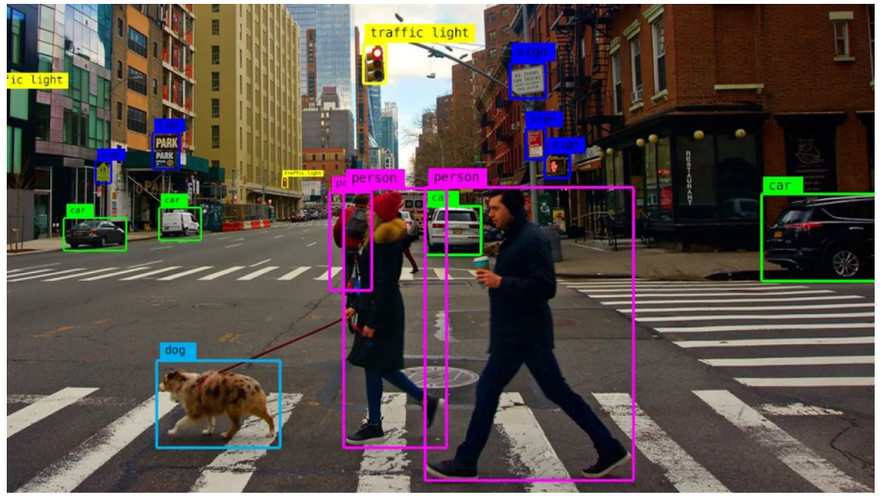
Power System

Rechargeable battery pack provides extended night operation, with charging stations at municipal depots

AI & Machine Learning

Computer Vision Pipeline

Our system employs a YOLO (You Only Look Once) or MobileNet architecture for real-time object detection. The model is trained on thousands of labeled images showing various types of street waste, from plastic bottles to food packaging. During operation, the camera captures 30 frames per second, which are processed sequentially. Detected waste items are classified by type and size, with confidence scores above 85% triggering collection protocols.



Route Optimization

Citizen reports and historical data feed into a reinforcement learning algorithm that continuously improves cleaning routes. The system prioritizes high-density waste areas while minimizing travel distance and energy consumption.

Citizen Reporting Platform

01

Photo Capture

Users take picture of waste location using smartphone camera

02

Location Tagging

GPS automatically marks coordinates where photo was taken

03

AI Verification

System validates waste type and seriousness before accepting report

04

Route Integration

Verified reports update cleaning routes for next night's patrol

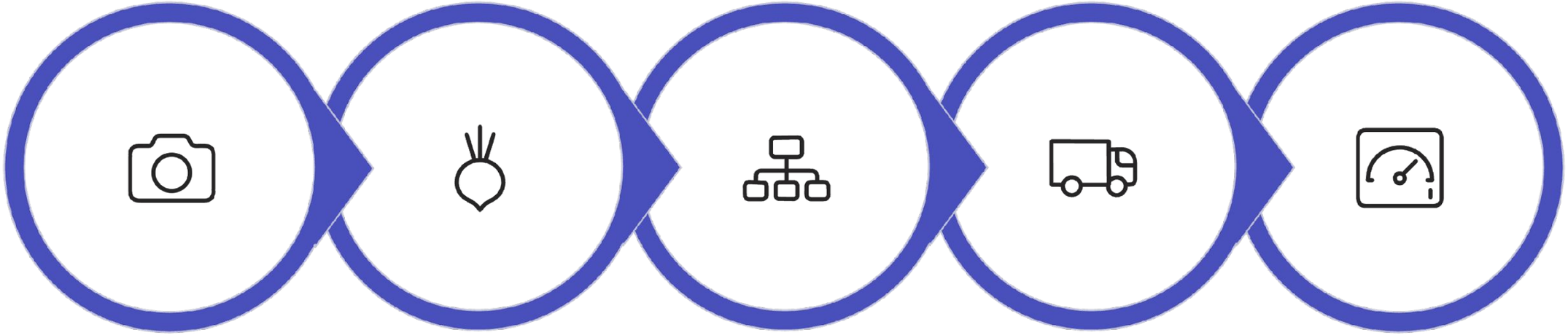
05

Status Updates

Users receive notifications when their reported area is cleaned

The web platform provides municipalities with a comprehensive heatmap dashboard showing waste concentration patterns, complaint resolution rates, and collection efficiency metrics.

System Workflow



Upload Photo

AI Verify

Optimize Route

Autonomous Clean

Update Dashboard

Implementation Roadmap

Phase 1: Prototype

Build scaled-down autonomous truck with basic waste detection and collection capabilities

1

Phase 2: AI Training

Collect and label dataset, train computer vision models, achieve 85%+ detection accuracy

2

Phase 3: Web Platform

Launch citizen reporting app with GPS tagging and status tracking features

3

Phase 4: Pilot Program

Deploy 3 trucks in one city district for 30-day trial with municipal partnership

4

Expected Impact

75%

Reduction in street waste

Based on pilot projections

60%

Lower manpower costs

Automation efficiency gains

4X

Faster response time

From report to cleanup

Target deployment: Municipal corporations, smart city initiatives, and government sanitation departments. Scalable architecture supports expansion from district-level to city-wide coverage.